

Address

Helsinki,
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Hugo LECOMTE

POST-DOC IN GEODESY

Geomatics Engineer

(01/12/2024)

Mail

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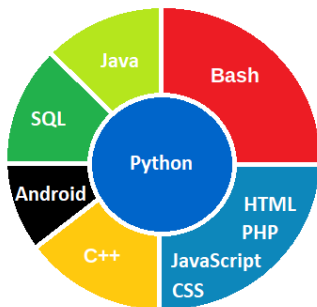
ORCID

0000-0002-7007-4748

Website

hulecom.github.io

Programming



OS Ability

Linux ★★★★★
Windows ★★★★★

Languages

French: Native language
English:

C1 with Europass guidelines
self-assessment
B2 TOEIC with 905 (10/2017)

Driving Licence

Full, clean

Degrees

- 2020 - 2023 PhD in geophysics [ITES, University of Strasbourg](#)
Obtained on the 25/10/2023. Directed by S. Rosat and M. Mandea
4 Rue Blaise Pascal, 67000 Strasbourg, FRANCE
- 2016 - 2019 Master's Degree in Geomatics Engineering [ENSG](#)
Geodesy, Remote sensing, Photogrammetry and Programming
and also Land surveying, GIS, Cartography
6 Av Blaise Pascal, 77420 Champs-sur-Marne, FRANCE

Professional Experience

- 08/25 - 08/28 Senior research scientist [FGI, National Land Survey of Finland](#)
- 11/23 - 06/25 Post-doctorant in geodesy / gravimetry [LEGOS / CNRS, Toulouse](#)
Study of regional and global mean sea-level and continental surface water from space gravimetry
Gravimetry - GRACE - Oceanography
- 11/20 - 10/23 PhD student in geodesy, geophysics [ITES, University of Strasbourg](#)
PhD title: Magnetic, gravimetric and geodetic constraints on the dynamics of the fluid core
Gravimetry - Geomagnetism - GRACE
- 10/19 - 10/20 Research and Development Engineer in remote sensing [IGN](#)
Creation and treatment of time series with Sentinel 1&2 images to classify crop type (Monitoring PAC),
Geodesy instruction (Geodetic reference system: Lecture courses)
Remote sensing - Geodesy - Database - Radar
- 05/19 - 09/19 End of study internship [Landmælingar Íslands, Iceland](#)
Connection of an old geodetic datum: Reykjavik local
Reference frame - GNSS tie - geodesy tools
- 05/18 - 08/18 Research internship [Paris Observatory, SYRTE team](#)
Determination of a high-resolution geopotential model in coastal areas with atomic clock measurements
Chronogeodesy - Geoid determination

Collaborations and Contributions

- 2023 - 2025 Participant of the SLBC_CCI project, Phase II [ESA CCI](#)
Magellium, TU Dresden, LEGOS, UBO-LOPS, Uni. Bristol, UNIBO, NERSC
Lead productions for a work-package (with deliverables for the project)
- 2023 - ... Development of a Python library **LENAPY** to process scientific dataset [LEGOS / CNES](#)
(Oceanography from S. Fourest and R. Guillaume-Castel / Gravimetry from H. Lecomte)
- 2023 - 2024 Reviewer for different manuscript submitted to
Communications Earth and Environment, JGR: Solid Earth, International Association of Geodesy Symposia, Advances in Geodesy and Geoinformation, Artificial Satellites

Publications

Total: 4 articles

- 2024 Gap filling between GRACE and GRACE-FO missions: assessment of interpolation techniques [10.1007/s00190-024-01917-3](https://doi.org/10.1007/s00190-024-01917-3) [Journal of Geodesy](#)
H. Lecomte, S. Rosat, M. Mandeia
- 2023 Uncertainty of Low-Degree Space Gravimetry Observations: Surface Processes Versus Earth's Core Signal [10.1029/2023GL104790](https://doi.org/10.1029/2023GL104790) [JGR: Solid Earth](#)
H. Lecomte, S. Rosat, M. Mandeia, J-P. Boy, J. Pfeffer
- 2023 Gravitational constraints on the Earth's core differential rotation [10.1029/2023JB026503](https://doi.org/10.1029/2023JB026503) [GRL](#)
H. Lecomte, S. Rosat, M. Mandeia, M. Dumberry

Supervision & Teaching

- 2024-25 2 Master Deg. internships co-supervision Comparison between altimetry and gravimetry [LEGOS](#)
- 2019-2024 General introduction to geodetic reference systems for Master 2 and 3rd year engineering students (PPMD) [ENSG](#)
- 2024 Space geodesy : Sensing changes on Earth system, for Master 1 students [Uni. Toulouse](#)
Geodetic reference frame and gravimetry
- 2020-2023 PhD additional education missions [EOST](#)
-Tutorial classes in space geodesy for 1st year engineering students (Radar & INSAR)
-Tutorial classes in geodesy for 2nd year engineering students (GNSS & reference frames)
-Tutoring classes in Mathematics for 1st year bachelor students
-Practical work in Python applied to geodesy for 1st year engineering students

Conferences

Total: 15 participations

(full list on personnel [website](#))

- 10/24 Level-3 CNES ensemble solution, assessment with lake altimetry (presentation) [GSTM](#)
H. Lecomte, A. Blazquez, B. Meyssignac, C. D'Hondt
- 10/24 Multi-decadal Satellite Gravity Mission Simulations to Resolve Long-term Signals in the Ocean (presentation) [GRACE Science Team Meeting](#)
M. Schlaak, H. Lecomte, R. Pail, B. Meyssignac, A. Blazquez
- 11/23 Detectability Criteria of the Earth's Core Processes with MAGIC Gravity Field Level 2 Products (poster) [MAGIC Science and Applications Workshop](#)
H. Lecomte, S. Rosat, M. Mandeia, M. Dumberry
- 07/23 GRACE/SLR-based Gravity Field and the Earth's Core: New Estimates for Parameters of Key Core Processes (invited talk) [IUGG](#)
H. Lecomte, S. Rosat, M. Mandeia, M. Dumberry
- 12/22 Observability of the Earth's core signals and geomagnetic jerks in GRACE-based gravity field. (poster) [AGU Fall Meeting](#)
H. Lecomte, S. Rosat, M. Mandeia, M. Dumberry
- 10/22 Comparison of gap-filling temporal methods to improve GRACE and GRACE-FO time series. (presentation) [GRACE Science Team Meeting](#)
H. Lecomte, S. Rosat, M. Mandeia
- 10/21 Comparison of SWARM and GRACE time-variable gravity field at low degree spherical harmonics. (poster) [Swarm Data Quality Workshop](#)
H. Lecomte, S. Rosat, M. Mandeia